

Polynomial Codes

- ⊕ They are also known as **CRC** codes
 - ➡ Check bits are generated in the form of a **Cyclic Redundancy Check**
 - ➡ Implemented using the **shift-register circuit**
- ⊕ The **k** information bits ($i_{k-1}, i_{k-2}, \dots, i_1, i_0$) are used as binary coefficients to form the information polynomial of degree **(k - 1)**:

$$i(x) = i_{k-1}x^{(k-1)} + i_{k-2}x^{(k-2)} + \dots + i_1x + i_0$$
- ⊕ The polynomial code uses **binary polynomial arithmetic** to calculate the codeword corresponding to the information polynomial

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Binary Polynomial Arithmetic

Addition: $(x^7 + x^6 + 1) + (x^6 + x^5) = x^7 + (1 + 1)x^6 + x^5 + 1 = x^7 + x^5 + 1$

Multiplication: $(x + 1)(x^2 + x + 1) = x^3 + x^2 + x + x^2 + x + 1 = x^3 + 1$

Division:

$$\begin{array}{r}
 \text{divisor } x^3 + x + 1 \overline{) \text{dividend } x^6 + x^5} \\
 \underline{x^6 + x^3} \\
 x^5 + x^4 + x^3 \\
 \underline{x^5 + x^3 + x^2} \\
 x^4 + x^2 \\
 \underline{x^4 + x^2 + x} \\
 x
 \end{array}$$

$x^3 + x^2 + x = q(x)$ **quotient**
 $x = r(x)$ **remainder**

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Polynomial Encoding

- ⊕ k information bits define the **information polynomial** of degree $(k - 1)$

$$i(x) = i_{k-1}x^{(k-1)} + i_{k-2}x^{(k-2)} + \dots + i_2x^2 + i_1x + i_0$$

- ⊕ A CRC code is specified by its **generator polynomial** of degree $(n - k)$ to generate $(n - k)$ check bits

$$g(x) = x^{(n-k)} + g_{n-k-1}x^{(n-k-1)} + \dots + g_2x^2 + g_1x + 1$$

- ⊕ $x^{(n-k)} i(x)$ is the **dividend polynomial**
- ⊕ Find the **remainder polynomial** $r(x)$ of at most degree $(n - k - 1)$

$$x^{(n-k)} i(x) = q(x) g(x) + r(x)$$

- ⊕ Get the **codeword polynomial** of degree $(n - 1)$

$$b(x) = x^{(n-k)} i(x) + r(x)$$

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E.g. $k=4, n-k=3, n=7$

$$g(x) = x^3 + x + 1$$

info bits: $(0 \ 0 \ 1)$

① $i(x) = 0 \cdot x^3 + 1 \cdot x^2 + 0 \cdot x + 1 \cdot x^0$
 $= x^2 + 1$

② $g(x) = x^3 + x + 1$

③ $a(x) = i(x) \cdot x^{n-k} = x^5 + x^3$

④ long division

$$\begin{array}{r} x^2 \\ x^3 + x + 1 \overline{) x^5 + x^3} \\ \underline{x^5 + x^3 + x^2} \\ x^2 = r(x) \end{array}$$

OR:

$$\begin{array}{r} 1 \\ 1011 \overline{) 0101000} \\ \underline{1011} \\ 100 \leftarrow \text{check bits} \end{array} \quad \Rightarrow \quad B: (0101100)$$

⑤ $B(x) = a(x) + r(x)$
 $= x^5 + x^3 + x^2$

⑥ $B:$
 $(0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0)$
 info bits check bits

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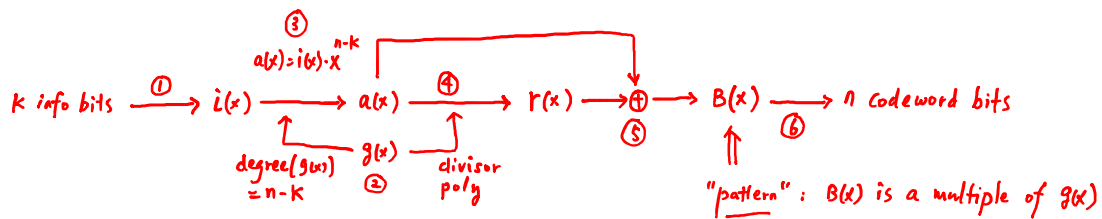
"Pattern"

$$\begin{aligned}
 B(x) &= \underline{a(x)} + r(x) \\
 &= \underline{g(x) * q(x)} + r(x) \\
 &= g(x) * q(x) + \underbrace{(r(x) + r(x))}_0 \\
 &= \underbrace{g(x)}_{\substack{\uparrow \\ \text{generator polynomial}}} * \underbrace{q(x)}_{\substack{\rightarrow \\ \text{quotient poly.}}}
 \end{aligned}$$

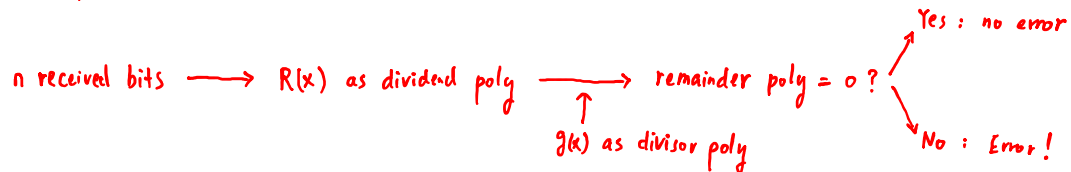
$\Rightarrow B(x)$ is a multiple of $g(x)$ "pattern"

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At encoder :



At decoder :



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The Pattern in Polynomial Code

- ⊕ All codeword polynomials satisfy the following **pattern**:

$$B(x) = x^{(n-k)} i(x) + r(x) = q(x)g(x) + r(x) + r(x) = q(x)g(x)$$

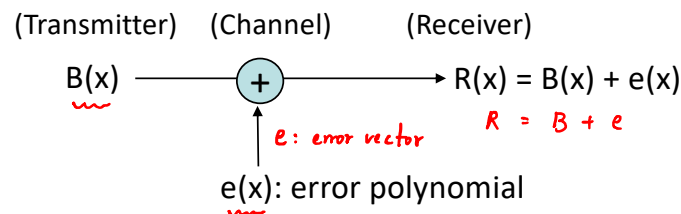
In other words, **all codeword polynomials are multiples of $g(x)$!**

- ⊕ Receiver should

- ➡ Convert the received n -bit block into a degree- $(n-1)$ dividend polynomial
- ➡ Divide the dividend polynomial by $g(x)$
- ➡ Check whether the remainder polynomial is zero
- ➡ If the remainder polynomial is non-zero, then the received n -bit block is not a valid codeword → error detected

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Undetectable Errors



- ⊕ $e(x)$ has “1” coefficients in error locations & “0” coefficients elsewhere

$e(x)$ is undetectable iff the corresponding $R(x) = B(x) + e(x)$ is still a multiple of $g(x)$

$$\begin{cases} R(x) = B(x) + e(x) = g(x) * q(x) + e(x) \\ R(x) = g(x) * q'(x) \end{cases}$$

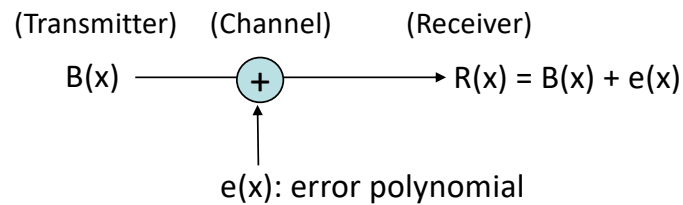
$$\Rightarrow g(x) * q'(x) = g(x) * q(x) + e(x)$$

$$\Rightarrow e(x) = g(x) * (q'(x) - q(x))$$

$$\Rightarrow \underline{e(x) \text{ is a multiple of } g(x)}$$

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Undetectable Errors



⊕ $e(x)$ has “1” coefficients in error locations & “0” coefficients elsewhere

⊕ If $e(x)$ is a multiple of $g(x)$, then:

$$R(x) = B(x) + e(x) = q(x)g(x) + q'(x)g(x) = [q(x) + q'(x)]g(x)$$

⇒ If a * non-zero error polynomial is divisible by $g(x)$,
then the corresponding error is undetectable

* $e(x)$ is detectable iff $e(x)$ cannot be divided by $g(x)$

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E.g. $g(x) = x^3 + x + 1$ $n-k=3$, $K=4$, $n=7$

$$e = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]_{1 \times 7} \quad \begin{cases} M=2 & (2\text{-bit error}) \\ L=7 & (\text{error burst of length } 7) \end{cases}$$

$\uparrow \qquad \qquad \qquad \uparrow$

$\Downarrow$

$$e(x) = x^6 + 1$$

$$\begin{array}{r} 1011 \\ 1011 \overline{) 1000001} \\ \underline{1011} \downarrow \downarrow \downarrow \\ 11001 \downarrow \\ \underline{1011} \downarrow \\ 1111 \\ \underline{1011} \\ 100 \\ \text{non-zero} \end{array}$$

⇒ $e(x)$ is not a multiple of $g(x)$

⇒ e is detectable

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E.g. $g(x) = x^3 + x + 1$ $K=4, n=7$

$$e = [0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0]_{1 \times 7} \quad \left\{ \begin{array}{l} M=1 \\ L=1 \end{array} \right. \quad \text{single bit error}$$

$\uparrow$
 $\uparrow$

$$\left. \begin{array}{l} e(x) = x^3 \quad (\text{single term}) \\ g(x) = x^3 + x + 1 \quad (\text{has more than 1 terms}) \end{array} \right\} \begin{array}{l} \text{single-term poly.} \\ \text{cannot be divided by } g(x) \end{array}$$

$\Rightarrow$ detectable

$$e = [0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1]_{1 \times 7} \quad \left\{ \begin{array}{l} M=3 \\ L=3 \end{array} \right.$$

$\uparrow \quad \uparrow$
 $\uparrow \quad \uparrow$

$$\left. \begin{array}{l} e(x) = x^2 + x + 1 \\ g(x) = x^3 + x + 1 \end{array} \right\} \begin{array}{l} \deg(e(x)) = 2 < 3 = \deg(g(x)) \\ \Rightarrow e(x) \text{ cannot be divided by } g(x) \\ \Rightarrow \text{detectable} \end{array}$$

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Error Burst:

$$e = [0 \ \dots \ 0 \ \overbrace{? \ ? \ \dots \ ?}^L \ 0 \ \dots \ 0]_{1 \times n}$$

$\uparrow \quad \quad \quad \uparrow$
 $i+L-1 \quad \quad \quad i$

$$\begin{aligned} e(x) &= 1 \cdot x^{i+L-1} + ? x^{i+L-2} + \dots + ? x^{i+1} + 1 \cdot x^i \\ &= \underbrace{x^i}_{\substack{\uparrow \\ \text{single-term}}} \cdot \underbrace{(x^{L-1} + ? x^{L-2} + \dots + ? x + 1)}_{\deg = L-1} \end{aligned}$$

when $\boxed{L-1 < \underbrace{n-K}_{\deg(g(x))}}$, $e(x)$ cannot be divided by $g(x)$

$\Rightarrow e$ is detectable

$$\left\{ \begin{array}{l} L-1 < n-K \\ L < n-K+1 \end{array} \right. \quad \begin{array}{l} \Rightarrow FUE(L) = 0 \text{ when } L-1 < n-K \\ \text{"short error bursts"} \end{array}$$

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